Unit 5

Measurement of Force, Torque and Pressure

(Handwritten Notes)

Syllabus Topics Covered

1. Force: direct methods - analytical balance, platform balance
2. Elastic members: load cells, cantilever beams, proving rings
3. Torque: torsion bar, servo-controlled and absorption dynamometers
4. Pressure: standards and calibration, basic methods
5. Dead weight gauges and manometers
6. High and low pressure measurement
7. Elastic pressure transducers

1. Measurement of Force

Force ($F = m\,a$, unit **newton N**) is a **vector** quantity that **changes the motion or shape** of a body. Methods of measuring force fall into two classes:

1. **Direct (balancing) methods** - **compare** the unknown force with a **known gravitational force** (known mass) or with a **magnitude** produced by a lever, for example **analytical balance, platform balance** .
2. **Indirect methods** - **measure the effect** of the force: **deflection of an elastic member** (spring) -> **load cell, proving ring** , **hydraulic / pneumatic** pressure balancing, **magnetic / piezo-electric** effects, **acceleration** of a known mass.

1.1 Direct method: Analytical balance

A **very precise equal-arm balance** used to measure **small masses** (0.1 mg precision) -> **weight = force** . Works on the **principle of the moment (lever)** : at balance, $W_1 \times L_1 = W_2 \times L_2$.

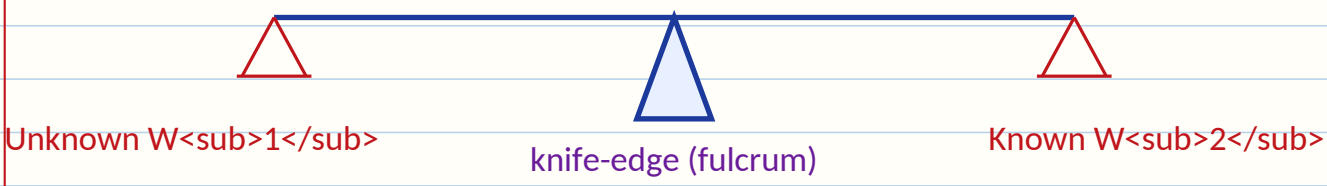


Fig 5.1: Equal-arm (analytical) balance

- **Construction** : **beam** (light, rigid, with **three agate knife edges** - central fulcrum, two outer for pans), **pointer** moving on a **scale** (sensitivity), **pans** , **beam arrest** , **levelling screws** , **glass case** (draught protection), **rider / chain** for fractional weights.
- **Equal arms** -> $W_1 = W_2$ independent of the arm length (if equal) - **double-weighing (Gauss)** method eliminates **arm-length inequality** .
- **Sensitivity** : S = **angular deflection per unit mass** = **(L) / (W_b h)** : increases with **longer arms** , **lighter beam** , and **CG nearer the fulcrum** (small h).
- **Modern** : **electronic (electromagnetic force-restoration)** balances; **resolution 1 microgram** .

1.2 Platform (weighing) balance

Used to weigh **heavy loads** (up to tonnes): the **load on a platform** is transmitted through a **system of levers** to a **beam** balanced by **small known weights** (poise), the ratio of lever arms (e.g. **1 : 10 : 100**) reduces the force.

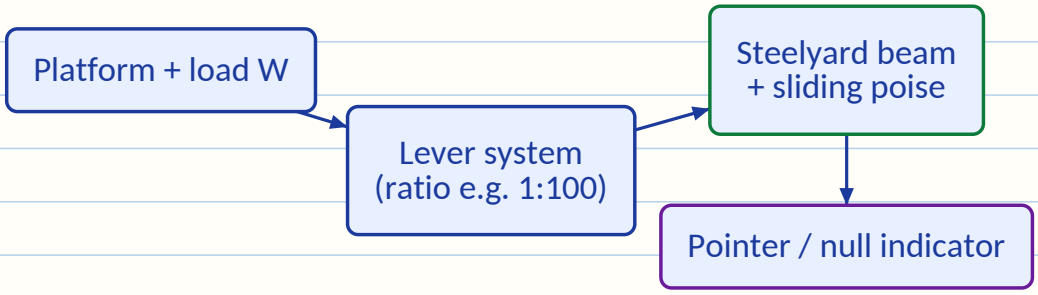


Fig 5.2: Platform balance (block diagram)

- **Principle** : load W acts on platform -> **main lever** -> **steelyard rod** -> **beam** ; the **poise** weight w is slid until the beam is **horizontal (null)** -> **W = w x lever ratio** .
- **Advantages of direct (balance) methods** : **high accuracy** , **independent of elasticity** , **simple** , **no calibration drift** . **Limitations** : **slow** , **not suitable for dynamic** force, **location-dependent g** (gravity) effect, **manual** .

2. Elastic Force Measuring Members

Force is applied to an **elastic member** (of known **spring constant**) and the **resulting deformation (strain / deflection)** is measured by a **displacement transducer / strain gauge** . The member's behaviour obeys **Hooke's law : F = k x** (within elastic limit).

1. **Spring balance** (helical): extension ∝ load; **simple, low accuracy** .
2. **Load cells** (**strain-gauge, hydraulic, pneumatic, piezo, magnetoelastic**).
3. **Cantilever beam** ; **proving ring** ; **elastic ring** ; **column** .

2.1 Load cells

A **load cell** is a transducer that converts **force (load) into an electrical signal** . The **elastic body** carries **strain gauges** wired in a **Wheatstone bridge** ; **output voltage ∝ force** .

Load cell type	Elastic element	Features
Column (compression/tension)	solid / hollow **steel cylinder** with 4 gauges: 2 axial, 2 transverse (Poisson)	high capacity (kN - MN); **very rigid** ; compensates bending
Cantilever (bending beam)	**beam** fixed at one end; load at free end; gauges **top & bottom**	**low capacity (up to ~ 5 kN)** ; high output; **weighing scales**
Shear beam	**web with shear strain** at 45 degrees	**insensitive to load position** ; platform scales
Ring / S-type	ring or S-shaped member	tension & compression; **medium capacity**
Hydraulic	**piston + diaphragm** with **oil** ; **pressure** ∝ force (P = F/A) read by a **gauge**	**no electric power** ; **hazardous areas** ; **very high capacity**
Pneumatic	diaphragm and **air back-pressure** (nozzle-flapper)	**clean, safe** ; **medium accuracy** ; food / chemical plants

Strain-gauge load cell

Load cell output = (V_exc x GF x ε) for full bridge; ε = F / (A E) (column); sensitivity expressed in mV/V (e.g. 2 mV/V at rated capacity).

- Accuracy 0.02 - 0.1% of rated load; temperature compensated .

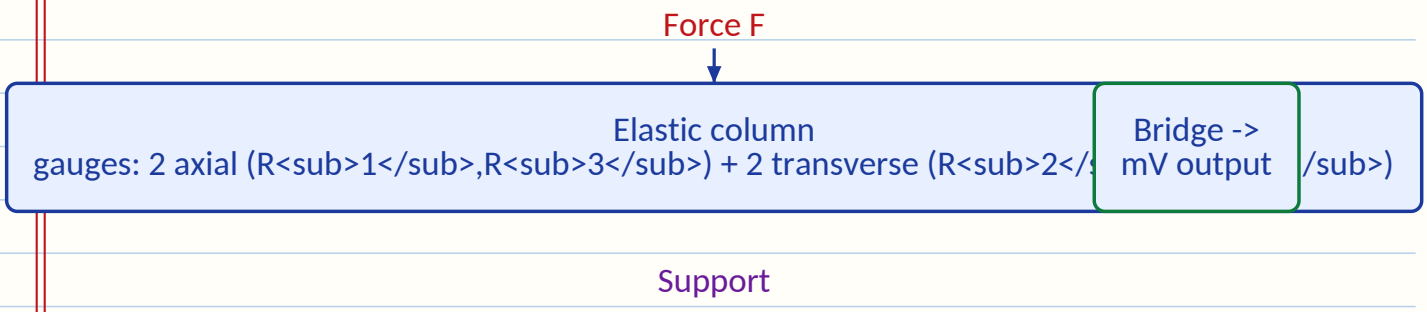


Fig 5.3: Strain-gauge load cell

- Advantages : wide range (g to MN) , compact , good accuracy , fast response , remote reading , used in industrial weighing , material testing machines , crane overload protection . Disadvantages : temperature/creep , overload damage , need for excitation, amplifier , side-load sensitivity .

2.2 Cantilever beam

A simple beam rigidly fixed at one end with the force applied at the free end . The deflection (or strain at the surface) is proportional to the force.

Cantilever relations

Deflection at free end: $\delta = F L^3 / (3 E I)$, with $I = b t^3 / 12$ (rectangular).

Surface strain near the root: $\varepsilon = 6 F L_g / (E b t^2)$ (L_g = distance from the load to gauge)

-> $F = \varepsilon E b t^2 / (6 L_g)$: top gauge in tension, bottom gauge in compression -> half/full bridge, doubled output, temperature-compensated.

- Example : steel beam b = 20 mm, t = 3 mm, L = 150 mm, E = 200 GPa, measured ε = 500 με: F = 500e-6 x 200e9 x 0.02 x $(0.003)^2$ / (6 x 0.15) = 20 N .
- Uses : low forces (weighing, dynamometer for machining forces), accelerometers (mass at the tip), micro-force sensors , AFM probes .

2.3 Proving ring

A proving ring is a steel ring (annular) elastic member with two diametrically opposite loading points ; the change in diameter (deflection) under load is measured with a micrometer / dial gauge / LVDT and calibrated to give the force .

Used as a **secondary standard** for **calibrating tensile & compression testing machines** (**load calibrations**).

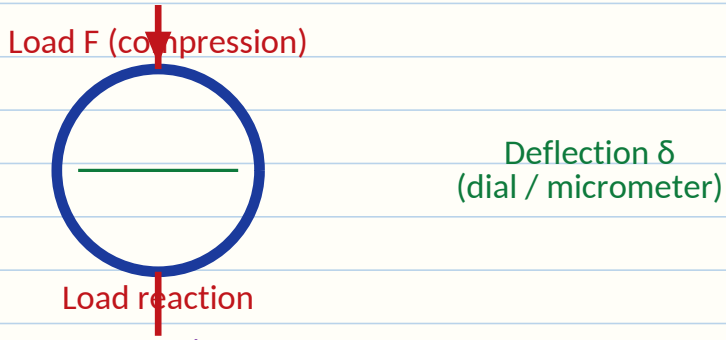


Fig 5.4: Proving ring

Deflection of proving ring

$\delta = 0.149\ F\ R^3 / (E\ I)$ (R = mean radius; $I = b\ t^3 / 12$; b = axial width, t = wall thickness)

- Linear response; **sensitivity increases with R** , decreases with t.

- **Features** : **simple, rugged** , **high accuracy (+-0.1%)** , **can be used in compression or tension** ; **vibrating-reed** between the diametral points (micrometer + reed resonance indicator) for **precise determination** of the diameter change; **temperature** corrections needed (**steel E changes**).
- **Capacity** : 2 kN - 2 MN; **calibration standard** as per **IS 1828 / ISO 376** .

Force measuring device	Principle	Typical use
Analytical / platform balance	direct, lever	mass / weights
Load cell	strain gauge on elastic body	industrial weighing, testing
Cantilever beam	bending strain/deflection	small forces
Proving ring	ring deflection	calibration of testing machines

3. Measurement of Torque

Torque (T = F x r) is the **twisting moment** acting on a shaft (N.m). It is measured by a **dynamometer** , and **power** is found from torque and speed:

Power transmitted

$P = T\omega = 2\pi N T / 60$ (watts, with T in N.m and N in rpm)

$kW = 2\pi N T / (60 \times 1000)$.

Class of dynamometer	Action	Examples
Absorption	absorb the power developed by the prime mover - **convert to heat** or electricity	**Prony brake** , **rope brake** , **hydraulic (water brake)** , **eddy-current** , **electric generator**
Driving	**drive** the machine and measure the **input** torque (**loading machines**)	electric motor dynamometer
Transmission	**transmit** power from engine to load, **measuring the torque in between**	**torsion bar** , belt, **epicyclic gear** , strain-gauge shaft

3.1 Torsion bar (transmission) dynamometer

The **power is transmitted through a shaft (torsion bar)** of **known torsional rigidity (G J)** ; the **angle of twist** θ between the two ends is **proportional to torque** .

Torsion

$T / J = G\theta / L = \tau / r \rightarrow T = G J \theta / L$ ($J = \pi d^4 / 32$ for a solid shaft; G = modulus of rigidity; L = gauge length).

- **Twist angle is small** - amplified by **optical lever** , **mechanical (dial) gauge** , **strain gauges (+-45 degree)** , **magnetic pickups** , **phase shift between two toothed discs (inductive)** , **slotted discs (stroboscopic / photoelectric)** .

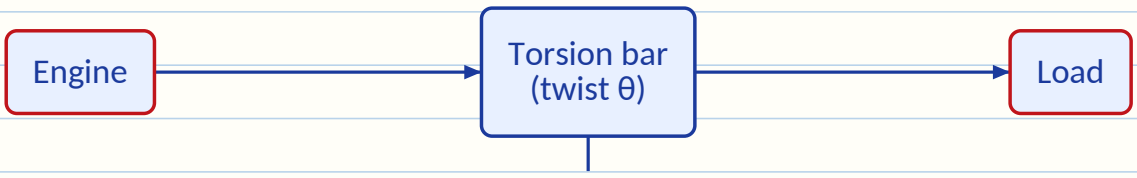


Fig 5.5: Torsion-bar dynamometer

- **Phase-difference (magnetic) type** : **two toothed wheels** at the ends of the bar with **magnetic pickups** : at no load, the pulses are **in phase** ; at torque the **time (phase) difference** $\Delta t \propto$ twist: $\theta = \omega \Delta t \rightarrow T = G J \omega \Delta t / L$; gives **torque & speed** on **rotating shaft** (contact-less, **no slip rings**).
- **Strain-gauge torque shaft** : **four gauges at 45 degrees** on shaft in **full bridge** ; signal by **slip rings / rotary transformer / telemetry** .
- **Advantages** : **measures running torque without absorbing the power** , **simple** , **good accuracy (0.25-1%)** . **Disadvantages** : **bar flexibility** may affect the system dynamics, **limited by shaft strength** , **signal transfer** problems.

3.2 Servo-controlled (electrical) dynamometer

A **cradled (trunnion-mounted) electrical machine** (DC / AC) whose **stator is free to rotate** a small angle: the **reaction torque on the stator = the torque on the shaft** . A **null-balance servo** automatically **restores** the stator position.

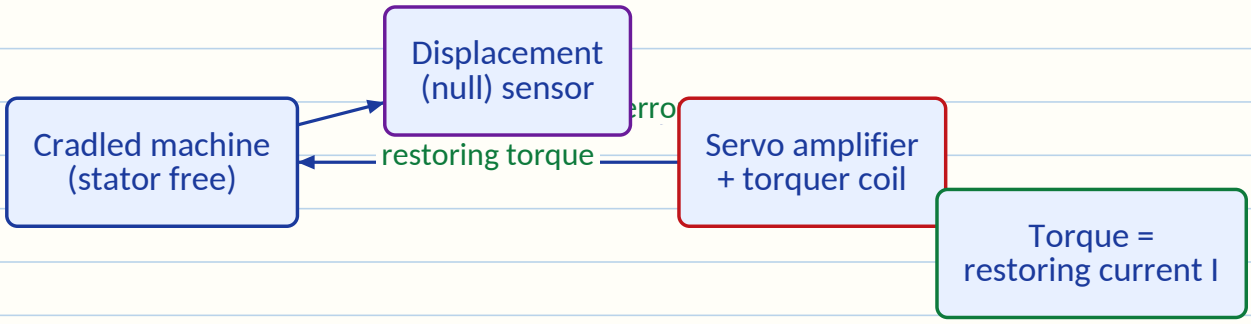


Fig 5.6: Servo-controlled dynamometer

- **Working** : when torque is applied, the **stator tends to rotate** ; the **sensor (LVDT, optical, capacitive)** detects the **displacement** ; the **servo amplifier** passes **current through a torque motor / coil** that **opposes** the movement till the **stator returns to null** ; **current I ∝ torque (T = K I)** ; read as **torque (N.m)** by a calibrated **ammeter** .
- **Advantages** : **no moving mechanical parts of measurement** - **high precision** , **fast** response, **continuous / automatic** , **can both absorb & drive (4-quadrant)** , works at **all speeds** , **remote and computer control** . **Disadvantages** : **costly** , **complex electronics** ; **limited power** range; **bearing friction** and **windage** corrections.

3.3 Absorption dynamometers

(a) Prony brake

- A **mechanical friction brake** : a **flywheel** (brake drum) rotating inside **wooden blocks** clamped by **bolts** , with a **lever arm** carrying a **dead weight W** (or a

spring balance) at a **distance L** from the axis; the **friction torque = W x L** (balance).

Prony brake

T = W L ; **Brake power BP = 2π N W L / 60** (W in N, L in m).

- Heat produced is removed by **water cooling** of the drum.

- **Disadvantages** : **unsteady** (coefficient of friction changes with **heat** , **self-energising** - jerks), **only for low power / speed** , **needs cooling** . **Advantages** : cheap, simple.

(b) Rope brake dynamometer

- A **rope (several turns)** wound round the **flywheel (drum)** ; one end carries **dead weight W** , the other a **spring balance S** ; the **difference** (W - S) is the **friction force** : **T = (W - S) (D + d) / 2** , where **D = drum diameter, d = rope diameter** .
- **BP = π N (W - S)(D + d) / 60** . Used for **small engines** ; **cooling water** in the drum; **rope wear** ; **unsteady** for large powers.

(c) Hydraulic (water-brake) dynamometer (Froude)

- A **rotor (disc with pockets)** rotates in a **casing (stator)** filled with **water** ; the **shear/vortex (churning) of water** absorbs power and **heats the water** ; the casing is **trunnion mounted** and the **reaction torque** is read through a **torque arm and a spring/dead weight** (**T = W L**). **Absorption controlled by water level / sluice plates** .
- **Advantages** : **high power & speed (MW)** , **compact** , **stable** , **continuous load control** . **Disadvantages** : **water supply / disposal** , **corrosion** , **cannot be run at low speed** , **braking torque ∝ N^2** .

(d) Eddy-current & electric dynamometers

- **Eddy-current** : a **toothed (or plain) rotor disc** turning in the **magnetic field of an excited stator coil** -> **eddy currents** in the **stator / rotor** produce **drag torque** , **power dissipated as heat** (water-cooled). **Torque ∝ excitation current** ; fast **response** , **no mechanical wear** , widely used for **engine testing** . **DC / AC generator type (cradled)** - **regenerative** , **absorbs or drives** .
- **Fan brake** , **air brake** : dissipate power by **fan** , for small / **high speed** machines (turbo).

4. Measurement of Pressure

Pressure = force per unit area ($P = F / A$) ; SI unit pascal ($Pa = N/m^2$) . Types:

- Absolute pressure - above perfect vacuum (zero reference). Gauge pressure - above atmospheric : P_abs = P_gauge + P_{atm} .
- Vacuum pressure - below atmospheric : P_abs = P_{atm} - P_vac. Differential pressure = difference between two pressures (P_1 - P_2).

Unit	Equivalent
1 bar	10^5 Pa = 100 kPa
1 atm (standard)	101.325 kPa = 760 mm Hg = 1.01325 bar
1 mm Hg (torr)	133.322 Pa
1 kgf/cm^2	98.07 kPa
1 psi	6.895 kPa

Standards and calibration

- Primary standards : dead-weight tester (piston gauge) and liquid-column manometers (mercury) - pressure derived from fundamental quantities (mass, length, g) ; accuracy 0.005 - 0.02% .
- Secondary standards : precision elastic gauges (test gauges, Bourdon / quartz), calibrated against primary ; working standards : industrial gauges and transducers .

1. Calibration procedure : connect the gauge under test to the dead-weight tester (hydraulic / pneumatic); apply known loads in steps upward (0, 20, 40 ... 100% of range) and downward ; record the gauge readings ; compute errors, hysteresis ; tap gently to overcome friction; apply corrections for local g , air buoyancy , temperature , head of oil (height difference); issue a calibration certificate .

Basic methods of pressure measurement

1. Balance (gravitational) methods : liquid column (manometer) , dead weight (piston) gauge .
2. Elastic methods : deformation of an elastic member - Bourdon tube, diaphragm, bellows, capsule .
3. Electrical methods : strain gauge, capacitive, piezo-electric, LVDT, potentiometric , resonant (vibrating wire) transducers.
4. Physical property methods (for vacuum): thermal conductivity (Pirani,

thermocouple) , ionization , viscosity .

5. Dead-Weight Gauges and Manometers

5.1 Dead-weight gauge (piston gauge tester)

Pressure is generated in a fluid (oil) by a piston-cylinder assembly of precisely known area A loaded with known weights W ; the pressure balanced is P = W / A (including the piston and carrier weight). Used to calibrate pressure gauges (up to 1000+ MPa).

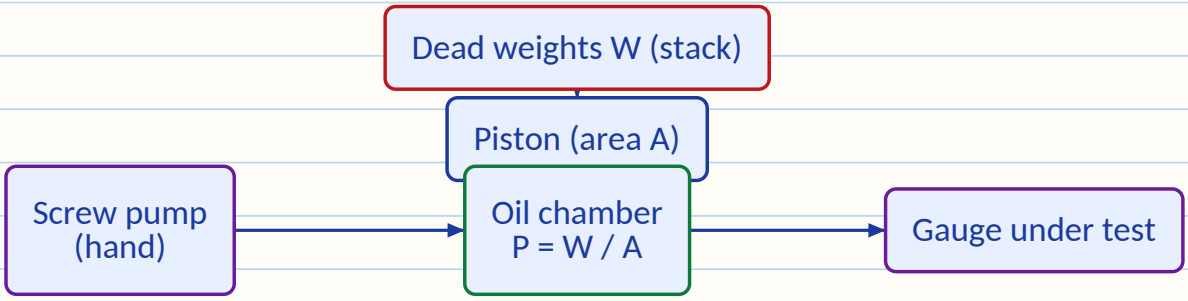


Fig 5.7: Dead-weight tester

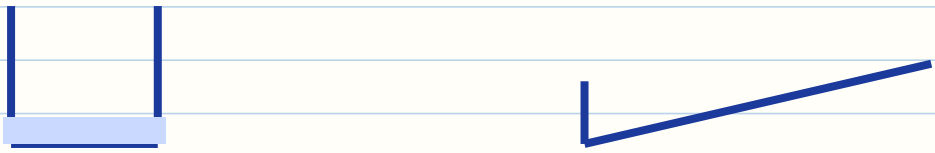
- Working : screw pump raises the oil pressure until the loaded piston floats (lifts slightly); the piston is rotated (to reduce static friction); then P = W / A equals the gauge pressure; the gauge under test is compared. Effective area A = (A_p + A_c) / 2 (piston and cylinder) and depends on pressure & temperature .
- Accuracy +-0.02% ; uses : calibration labs , standard for pressure transducers . Pneumatic version for low pressure (gas lubricated piston).
- Corrections : local g , air buoyancy , temperature of piston , height difference between piston reference level and gauge : ΔP = ρ g h .

5.2 Manometers (liquid-column gauges)

Measure pressure by balancing it with the height of a liquid column (Mercury, water, alcohol or coloured oil): P = ρ g h .

Manometer	Principle / equation	Notes
U-tube	$P_1 - P_2 = \rho g h$ (h = difference in levels)	simple , range up to ~ 2 bar (Hg), both gauge & differential; zero setting
Well (cistern) type	$P = \rho g h (1 + A_t / A_w)$; with large well area, single reading only on the tube	easier reading; barometer (Fortin)

Inclined tube	P = ρ g L sin θ : h magnified by 1 / sin θ	small pressures (draft gauges) ; sensitivity increases as θ decreases (typically 10 degree)
Micro-manometer	optical / micrometer reading of the level	very small pressure differences (0.01 mm of water)



(a) U-tube (b) Inclined tube

- Advantages : simple, cheap, accurate , no calibration needed (primary) .
  Disadvantages : bulky, fragile , not for high pressure or dynamic , temperature and capillarity effects (correction $\Delta h = 4\sigma \cos\theta / (\rho g d)$ - use tubes > 12 mm), meniscus reading, mercury toxicity .

6. High and Low Pressure Measurement

6.1 High-pressure measurement

For pressures above ~ 1000 bar (100 MPa) , Bourdon gauges are replaced by electrical (resistance) methods.

- Bridgman gauge (manganin wire gauge): a coil of manganin wire (alloy of 84% Cu, 12% Mn, 4% Ni) is subjected to the fluid pressure inside a pressure vessel ; its resistance changes linearly with pressure ($\Delta R / R = K P$, $K \approx 2.3 \times 10^{-6}$ per bar , i.e. pressure coefficient ≈ 2.5×10^{-11} per Pa); read by a Wheatstone bridge ; range up to 1 - 10 GPa .

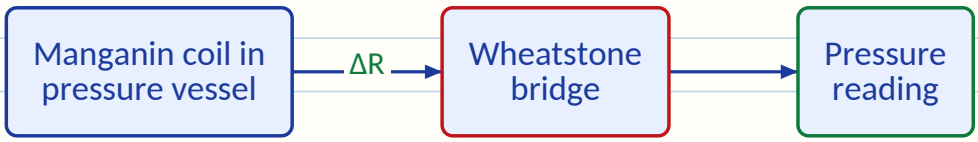


Fig 5.8: Bridgman gauge

- Features of manganin : high & stable pressure coefficient , very low temperature coefficient of resistance; hysteresis small ; calibrated against a dead weight tester . Used for explosive / shock, hydraulics & materials research .
- Other : strain-gauge (thick-diaphragm) transducers , heavy-walled Bourdon tubes , piezo-electric (dynamic), Piston gauge (to 1 GPa) .

6.2 Low-pressure (vacuum) measurement

Vacuum ranges: rough (1 - 760 torr) , medium (10^{-3} - 1 torr) , high (10^{-6} - 10^{-3}) , ultra-high (< 10^{-9} torr) . Units: torr, micron (1 micron = 10^{-3} torr) .

Gauge	Principle	Range (torr)
McLeod gauge	Boyle's law (compress a sample)	10^{-6} - 10
Pirani gauge	thermal conductivity (resistance of hot wire)	10^{-4} - 1
Thermocouple gauge	temperature of heated filament by thermocouple	10^{-3} - 1
Ionization gauge (hot cathode / Bayard-Alpert)	ion current ∝ gas density	10^{-10} - 10^{-3}
Knudsen / viscosity gauge	molecular momentum transfer	10^{-7} - 10^{-3}
Capacitance diaphragm	diaphragm deflection with capacitance	10^{-4} - 10^{3}

McLeod gauge

A primary standard for vacuum: it compresses a known large volume V of low-pressure gas into a small capillary volume (by raising mercury), then the pressure of the compressed gas is determined from the height h of mercury; Boyle's law gives the original pressure .

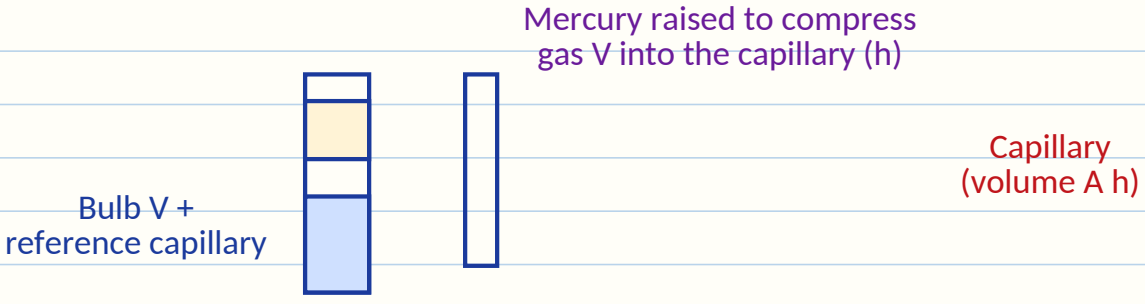


Fig 5.9: McLeod gauge

McLeod gauge equation

$P_1 V = P_2 (A h)$ with $P_2 = P_1 + h$ (mm Hg)

$P_1 = A h^2 / (V - A h) \approx A h^2 / V$ (square-law scale)

- A = capillary cross-section area, h = mercury height difference , V = volume of bulb + capillary.

- Advantages : absolute (independent of gas type) for permanent gases , calibration standard . Disadvantages : intermittent , not for condensable vapours (water vapour), mercury hazards, no continuous reading .

Pirani gauge

- A heated filament (tungsten / platinum) in the vacuum; heat loss by gas conduction depends on pressure (thermal conductivity ∝ P at low pressures). The filament resistance (temperature) changes and is measured by Wheatstone bridge (constant-temperature mode). Continuous , fast , range 10^{-4} - 1 torr ; gas-dependent calibration .

Ionization gauge

- Hot-cathode : electrons emitted by a heated filament ionize the gas molecules; ions collected at a negative collector give a current $I_c = S I_e P$ (S = sensitivity, I_e = emission current). Bayard-Alpert design with thin wire collector (reduces X-ray effect) reads down to 10^{-10} torr ; cold-cathode (Penning) type uses magnetic field and discharge ; fragile filament , gas-dependent .

7. Elastic Pressure Transducers

Pressure acts on an elastic member , which deforms (deflection ∝ pressure); the deflection is indicated mechanically (pointer) or sensed electrically (strain gauge, LVDT, capacitive, potentiometer).

7.1 Bourdon tube

A curved, flattened (elliptical cross-section) metal tube (C-shape , spiral , helical or twisted) closed at one end ; the open end is fixed to the pressure connection. Under internal pressure the cross-section tends to become circular , so the tube straightens (uncoils) - free end moves .

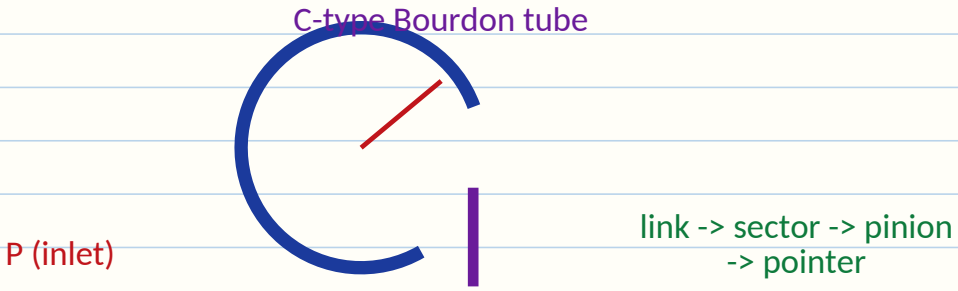


Fig 5.10: Bourdon pressure gauge

- Movement of the free end -> link -> quadrant (sector) -> pinion -> pointer (amplification of 1000 or more); hair spring removes backlash.
- Materials : phosphor bronze, brass, beryllium copper (low pressure); alloy steel / stainless steel (high pressure up to 7000 bar).
- Types : C-type (0 - 700 bar), spiral (more sensitive/ low pressure), helical (larger tip travel -> recorder, up to 7000 bar), twisted (for very high pressure / electrical pick-up).
- Advantages : simple, cheap, rugged, wide range (1 - 7000 bar) ; Disadvantages : hysteresis , not for fast-changing pressure , vibration damages; temperature effect; low-pressure limit ~ 0.5 bar (use diaphragm).

7.2 Diaphragm

A thin flexible disc (flat, corrugated, or dished) clamped at the edge ; pressure difference across it causes deflection of the centre (linear for small deflection: $y = 3 (1 - \nu^2) P R^4 / (16 E t^3)$).

- Flat (metal , linear for small deflection), corrugated (larger linear deflection , use for low pressure), slack diaphragm (rubber/fabric + spring, very low pressure: draft gauges); sensing : strain gauges bonded on the diaphragm (full bridge), LVDT , capacitive (variable gap) , piezo-resistive silicon (MEMS).
- Advantages : small, fast response (high natural frequency) , good for low and medium pressures (0.01 - 1000 bar) , differential pressure cells (DP transmitters, flow measurement). Disadvantages : limited deflection , temperature , non-linearity for large deflection.

7.3 Bellows and capsule

- Bellows : corrugated metal tube (brass, phosphor bronze, stainless steel) that expands/contracts axially with pressure against a spring : deflection $x = P A / k$ (A = effective area, k = spring rate). Used for low pressure (0.01 - 10 bar) , differential pressure (two bellows), pressure switches , aircraft altimeters .

- **Capsule (aneroid): two corrugated diaphragms joined at the edges** (disc-shaped chamber); **large deflection, very sensitive** ; stacked capsules **sum** the motion - used in **barometers (aneroid)** , **altimeters** , **vacuum gauges** .

Element	Typical range	Use
Bourdon tube	1 - 7000 bar	general purpose gauges
Diaphragm	0.01 - 1000 bar	low/medium, transmitters, DP
Bellows	0.01 - 10 bar	low pressure, switches, DP
Capsule	mbar - 1 bar	barometers, altimeters, vacuum

Quick Revision / Important Questions

1. Direct force measurement: analytical and platform balance; load cell, cantilever, proving ring.
2. Torque: torsion bar, servo-controlled and absorption dynamometers (Prony, rope brake).
3. Pressure standards, dead-weight gauge; manometers (U-tube, inclined).
4. High pressure (Bridgman) and low pressure (McLeod, Pirani, ionization).
5. Elastic pressure transducers: Bourdon tube, diaphragm, bellows, capsule.